

Name:

MATH 241

SUMMER 2014

QUIZ #8

Problem 1.

(a) (4 Points) Let $y[k]$ be the discrete convolution of $x[k]$ and $h[k]$ and $y[k]$ is given as

$$y[k] = h[k] * x[k] = \sum_{m=0}^k h[m]x[k-m]$$

Let $x[k] = h[k] = 0$ for $k < 0$, show that $Y(z) = X(z)H(z)$.

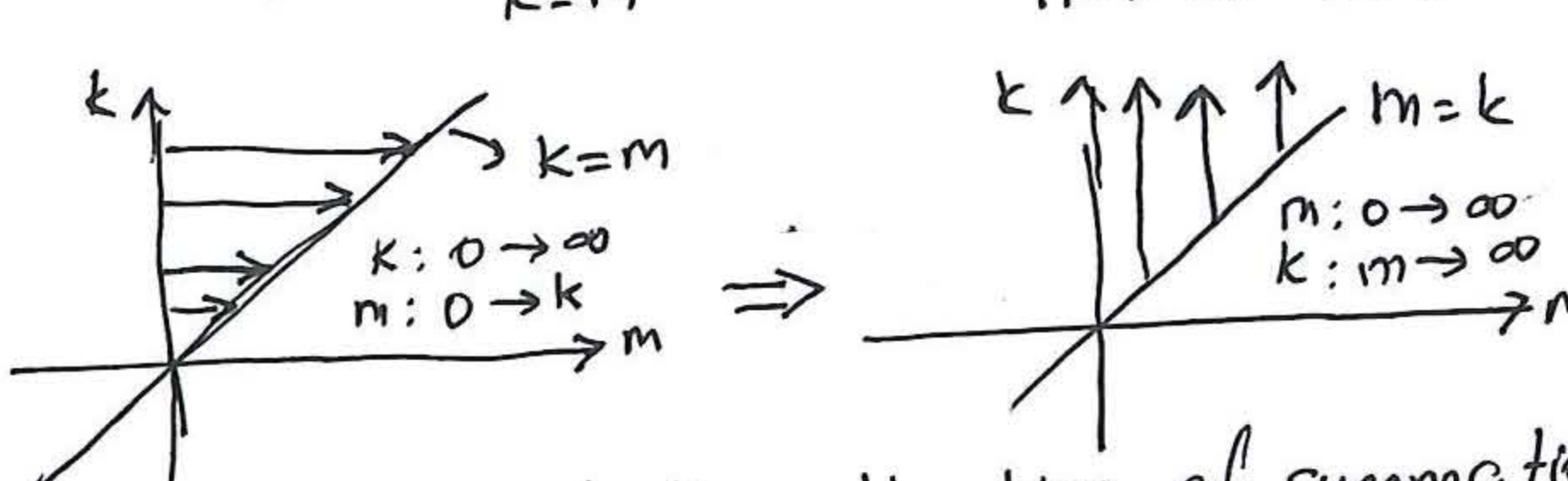
(b) Let $h[k] = 2^{-k}$ and $x[k] = 3^{-k}$ for $k \geq 0$ and $h[k] = x[k] = 0$ for $k < 0$.

(i) (3 Points) Find $X(z)$ and $H(z)$ then compute $Y(z) = X(z)H(z)$. Take the inverse Z transform of $Y(z)$ in order find $y[k]$.

(ii) (3 Points) Compute $y[k] = \sum_{m=0}^k h[m]x[k-m]$ and comment on the result.

Hint: $\sum_{m=k+1}^{\infty} h[m]x[k-m] = \sum_{m=-\infty}^{-1} h[m]x[k-m] = 0$ since $x[k] = h[k] = 0$ for $k < 0$.

(a) $Y(z) = \sum_{k=0}^{\infty} \sum_{m=0}^k h[m]x[k-m] z^{-k}$
 Step ① $= \sum_{m=0}^{\infty} \sum_{k=m}^{\infty} h[m]x[k-m] z^{-k}$
 $= \sum_{m=0}^{\infty} h[m] \sum_{k=m}^{\infty} x[k-m] z^{-k}$
 $= \sum_{m=0}^{\infty} h[m] \sum_{n=0}^{\infty} x[n] z^{-(n+m)} = \underbrace{\sum_{m=0}^{\infty} h[m] z^{-m}}_{H(z)} \underbrace{\sum_{n=0}^{\infty} x[n] z^{-n}}_{X(z)}$

Step ① 
 convert the direction of summation, we do summation in the same region!

(b) $X(z) = \frac{1}{1 - \frac{1}{3}z^{-1}}$ $H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$
 $Y(z) = \frac{1}{(1 - \frac{1}{3}z^{-1})(1 - \frac{1}{2}z^{-1})} = \frac{3}{1 - \frac{1}{2}z^{-1}} - \frac{2}{1 - \frac{1}{3}z^{-1}} \Rightarrow 3 \left(\frac{1}{2}\right)^k - 2 \left(\frac{1}{3}\right)^k$

$y[k] = \sum_{m=0}^k 2^{-m} 3^{-(k-m)} = 3^{-k} \sum_{m=0}^k \left(\frac{3}{2}\right)^m = 3^{-k} \frac{\left(\left(\frac{3}{2}\right)^{k+1} - 1\right)}{\frac{3}{2} - 1} = 2 \left(\frac{3}{2} \left(\frac{1}{2}\right)^k - 3^{-k}\right)$
 $= 3 \left(\frac{1}{2}\right)^k - 2 \left(\frac{1}{3}\right)^k \checkmark$

The results are the same!

Problem 2.

Consider the system of linear differential equations in the form of $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f}$ below,

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ f(t) \end{bmatrix} \text{ with the initial condition}$$

$$\mathbf{x}(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ and } f(t) = \begin{cases} e^{-2t} & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

- (7 Points) Find the solution of $x(t)$ for $t \geq 0$.
- (3 Points) Identify the homogeneous (zero input; $\mathbf{x}_{ZI}(t)$) and the forced (zero state; $\mathbf{x}_{ZS}(t)$) parts of the solution.
- (3 Points) Find the state transition matrix such that $\mathbf{x}_{ZI}(t) = \Phi(t; 0)\mathbf{x}(0)$. Verify that $\mathbf{x}_{ZI}(0) = \mathbf{x}(0)$ and $\mathbf{x}_{ZS}(0) = \mathbf{0}$.
- (2 Points) State and justify whether the system is stable or not.

$$X(s) = \underbrace{(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}(0)}_{\text{Homogeneous}} + \underbrace{(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{F}(s)}_{\text{forced}}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{bmatrix} s & -1 \\ 6 & s+5 \end{bmatrix}^{-1} = \frac{1}{s^2 + 5s + 6} \begin{bmatrix} s+5 & 1 \\ -6 & s \end{bmatrix}$$

$$\Phi(t; 0) = \mathcal{L}^{-1}((s\mathbf{I} - \mathbf{A})^{-1}) = \mathcal{L}^{-1}\left(\frac{1}{s+2} \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} + \frac{1}{s+3} \begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix}\right)$$

$$= e^{-2t} \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} + e^{-3t} \begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix}$$

$$\mathbf{x}_{ZI}(t) = \Phi(t; 0)\mathbf{x}(0) = \begin{bmatrix} -2e^{-2t} + e^{-3t} \\ 4e^{-2t} - 3e^{-3t} \end{bmatrix} \quad \mathbf{x}_{ZI}(0) = \begin{bmatrix} -2+1 \\ 4-3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \checkmark$$

$$\mathbf{x}_{ZS}(s) = \underbrace{(s\mathbf{I} - \mathbf{A})^{-1}}_{(s\mathbf{I} - \mathbf{A})^{-1}} \underbrace{\begin{bmatrix} 0 \\ 1/s+2 \end{bmatrix}}_{\mathbf{F}(s)} = \frac{1}{(s+2)(s+3)} \begin{bmatrix} 1/s+2 \\ s/s+2 \end{bmatrix} = \frac{1}{(s+2)^2(s+3)} \begin{bmatrix} 1 \\ s \end{bmatrix}$$

$$\mathbf{x}_{ZS}(s) = \frac{1}{(s+2)^2} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \frac{1}{(s+2)} \begin{bmatrix} A \\ B \end{bmatrix} + \frac{1}{s+3} \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ -6 \end{bmatrix} + 6 \begin{bmatrix} A \\ B \end{bmatrix} + \begin{bmatrix} 4 \\ -12 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow 6 \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} -6 \\ 18 \end{bmatrix} \Rightarrow \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

The constant numerator terms after obtaining common denominators.

$$\mathbf{x}_{ZS}(t) = t e^{-2t} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$+ e^{-2t} \begin{bmatrix} -1 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{-3t}$$

$$\mathbf{x}_{ZS}(0) = 0 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ -3 \end{bmatrix} = \mathbf{0} \checkmark$$

System is stable, since all roots are on the left half complex plane!